KHIZER HAYAT

Software Engineer | AI/ML Engineer | MLOps Engineer

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PROFESSIONAL SUMMARY

Software Engineering graduate (MUST, Class of 2026) with hands-on experience building end-to-end AI applications across machine learning, deep learning, computer vision, and backend engineering. Developed projects in skin-image analysis, model benchmarking, text-to-SQL, and automated video generation using Python, FastAPI, PyTorch, and TensorFlow. Experienced with Docker and Ollama for containerised and local model workflows, and currently deepening expertise in LLMs, Retrieval-Augmented Generation (RAG), LangGraph, and agentic AI systems.

TECHNICAL SKILLS

Programming Languages: Python, Java, C#
Machine Learning & Deep Learning: Supervised & Unsupervised Learning, Deep Learning, Neural Networks (CNNs), Computer Vision, Model Training & Evaluation
ML Frameworks & Data Science: PyTorch, TensorFlow/Keras, Scikit-learn, NumPy, Pandas, Matplotlib
MLOps & Model Tooling: Docker, Kubernetes (Foundational), Ollama, Model Benchmarking, Performance Profiling
Generative & Agentic AI (Learning): LLMs, RAG, LangGraph, AI Agents, Agentic Workflows
Web Development: FastAPI, RESTful APIs, JavaScript, React (Basic), Tailwind CSS
Databases: SQL, PostgreSQL, SQLite
Computer Vision & Media: OpenCV, OCR (Tesseract), Hugging Face Transformers, Text-to-Speech (TTS), FFmpeg
Core Computer Science: Data Structures and Algorithms (Python, Java)
Cybersecurity: Burp Suite, Linux Fundamentals, Network Scanning, OWASP Top 10

WORK EXPERIENCE

Cybersecurity Intern | Moboroid Software Company, Mirpur, AJK *Jul 2025 – Oct 2025*

- Performed network scanning and introductory penetration testing to identify security weaknesses.
- Analysed common web application vulnerabilities and their mitigation strategies with reference to the OWASP Top 10.
- Performed web security testing using Burp Suite and strengthened Linux command-line skills for security operations.

PROJECTS

Skyntri — AI-Based Skin Health Assistant *(Final Year Project)*
Technologies: Python, TensorFlow/Keras, Deep Learning (CNN), FastAPI, React, Tailwind CSS, Tesseract OCR, SQLite

- Trained a multi-label CNN-based deep learning model to classify ten visible skin-pattern classes from images.
- Built the React frontend and FastAPI backend with authentication, an AI chat assistant, progress tracking, and PDF reports.
- Implemented an OCR ingredient scanner with fuzzy matching to deliver skin-type-aware ingredient safety guidance.

Local AI Lab — AI Model Benchmarking Platform
Technologies: Python, Scikit-learn, PyTorch, Ollama, FastAPI, Docker

- Developed a platform to compare classical machine-learning and deep-learning models side by side.
- Evaluated models on accuracy, F1-score, inference latency, and resource utilisation to support model-selection decisions.
- Implemented experiment tracking and extended benchmarking to locally hosted LLMs through Ollama.

DataTalk — AI Database Analyst
Technologies: Python, LLMs, Scikit-learn, PostgreSQL, FastAPI, React

- Developing a text-to-SQL application that translates natural-language questions into executable PostgreSQL queries.
- Implementing query-intent classification, SQL validation for safe execution, and automated visualisation of query results.

ReelForge — AI Video Generation Pipeline *In Progress*
Technologies: Python, PyTorch, Transformers, OpenCV, TTS, FFmpeg, FastAPI

- Building an end-to-end pipeline that automates script processing, voiceover generation, subtitles, and video assembly.
- Integrating speech-recognition models and computer-vision-based scene classification for automated video editing.

EDUCATION

BS Software Engineering | Mirpur University of Science & Technology (MUST) | CGPA: 3.45/4.00 *Oct 2022 – Oct 2026*
Intermediate (HSSC) | Dr. A. Q. Khan School System and College, Bhimber, AJK | 76.73% *2019 – 2021*
Matriculation (SSC) | Jabbi National Foundation College, Bhimber, AJK | 84.91% *2017 – 2019*

LANGUAGES

Urdu | English